

Bolun ZHANG

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Address: 800 Dongchuan Road, Minhang District, Shanghai, 200240, China

Research Interests: Reinforcement learning and large language models.

PERSONAL PROFILE

A highly motivated individual with a passion for continuous learning and creative problem-solving. Has developed the ability to quickly master new concepts, write code, and optimise algorithms in areas such as autonomous driving, DDQN networks, and few-shot learning through professional competitions. Committed to advancing expertise in AI and contributing to innovative solutions that push the field forward.

EDUCATION

Shanghai Jiao Tong University

Shanghai, China

Bachelor of Engineering in Intelligent Perception

Sep 2022 – Jun 2026

- **GPA:** 3.51/4.3
- **Modules include:** Introduction to Engineering (93/100), Data Structure (95/100), University Physics (A) III (94/100), Fundamental of Analog Circuits (94/100), Science and Technology Innovation (Part 3-C) (95/100), Navigation and Positioning (95/100), Theory and Applications of artificial intelligence (93/100), Discrete-Time Signal Processing (90.20/100), Artificial intelligence algorithm practice (92.32/100), Modern Sensor Technologies (91.40/100)

PUBLICATION

Zhang, B., Zhuang, A., Liu, Y., Han, Z., Sun, F., Chen, L., Pan, Z., Guo, J., Xu, W., Shen, Z., Song, L., Li, Z., Zhang, Z., Dan, Y., Chen, Y., Sun, W., & Ji, Y. (2024) (2024). *Drone-Assisted Multi-Sensory Autonomous Driving System* [SCI]. In *Proceedings of the 31st CHINA SAE CONGRESS & EXHIBITION (SAECCE 2024)*.

- The system operates to support autonomous vehicles recognise blind spots and prevent obstacles through the use of unmanned drone technology. It resolves the safety concerns caused by autonomous vehicle LiDAR-based sensing when operated without assistance. It also incorporates artificial intelligence techniques such as YOLO (You Only Look Once) for object detection and DDQN (Double Deep Q-Network) reinforcement learning.

Fu, L., Guan, H., **Zhang, B.**, Yuan, H., Zhu, Y., Xu, J., Wang, Z., Qiu, L., Cai, X., Cao, X., Liu, W., Zhang, W., & Yu, Y. *CoreCodeBench: A Configurable Multi-Scenario Repository-Level Benchmark*

- CorePipe and CoreCodeBench address challenges in evaluating LLMs on engineering-level code. CorePipe automates repository-to-test conversion, generating three core code questions. CoreCodeBench offers configurable multi-scenario benchmarks to assess LLM applicability in engineering projects and reveal performance variations. (Submission to ICLR 2026 is in progress)

Fu, L., **Zhang, B.**, Guan, H., Zhu, Y., Qiu, L., Liu, W., Cao, X., Cai, X., Zhang, W., Yu, Y.

Automatically Benchmarking LLM Code Agents through Agent-Driven Annotation and Evaluation

- We propose an agent-driven benchmark pipeline using human supervision to generate diverse, challenging project-level tasks. We introduce PRDBench: 50 real-world Python projects across 20 domains, each with PRDs, criteria, and references. Using Agent-as-a-Judge, we score outputs beyond unit tests, assessing both code agents and evaluation agents effectively. (Submission to AAMAS 2026 is in progress)

ACADEMIC PROJECTS

Research on Large Language Models in Apex Lab

Nov 2024 - Present

Research Assistant

- Research and analyse current large language models and their implementation papers.
- Drill down into reinforcement learning as foundational knowledge for experiments.
- Comprehend the PRM (Process Reward Model) used in current large language models.
- Engage in the experimental work related to the implementation of large language models in the lab.

Research on Few-shot Learning

Jun 2024 – Nov 2024

Primary Researcher

- Studied the foundational framework of PyTorch.
- Conducted research on relevant literature and algorithms in few-shot learning.

- Reproduced and implemented few-shot learning frameworks using PyTorch on required datasets.

The 27th Student Innovation Practice Program of Shanghai Jiao Tong University

Apr 2023 – Oct 2024

Lead Principal Investigator

- Published an SCI paper on vehicle-machine collaboration: *Drone-Assisted Multi-Sensory Autonomous Driving System* (First Author). Please refer to the publication section for more details.
- Learned the basics of ROS (Robot Operating System) and independently completed a two-vehicle overtaking simulation.
- Developed a Deep Q-Network (DDQN) framework for reinforcement learning and trained an autonomous driving model within a self-drawn map.
- Gained in-depth knowledge of Gazebo simulation architecture, YOLOv8 for object detection, and drone APIs, while developing autonomous driving system frameworks and vehicle-machine collaboration techniques to ensure seamless integration of key components.
- Demonstrated strong leadership skills by guiding the research team, managing project timelines, and coordinating collaboration among interdisciplinary team members.

PROFESSIONAL EXPERIENCES

Meituan (Beijing)

Shanghai, China

Large Language Models Intern

Dec 2024 – Jun 2025

- Contributed to developing a pipeline for training large language models focused on code generation, using tools like Python, TensorFlow, PyTorch, and Hugging Face to improve model training and optimisation.
- Extensively used various large models, such as OpenAI O1, DeepSeek R1, and Claude, to construct and annotate datasets for AI performance evaluation.
- Participated in building AI agents and setting up development pipelines, including designing and implementing workflows to streamline the training, testing, and deployment of models.

AWARDS AND HONOURS

- **Intel Special Award & First Prize** - China-US Young Maker Competition, 2023
- **Second Prize** - National Undergraduate Student Electronics Design Contest, Shanghai Region
- **Second Prize** - Contemporary Undergraduate Mathematical Contest in Modeling, Shanghai Region

SKILLS & INTERESTS

- **Languages:** Chinese (native), English (proficient, IELTS Overall Band Score 7.0)
- **Programming Skills:** Python, C++
- **Tools & Frameworks:** ROS, YOLOv8, Gazebo, PyTorch
- **Other:** Advanced Excel (Data Processing)

LEADERSHIP AND ORGANISATIONAL CONTRIBUTIONS

**Youth League Committee, School of Electronic Information and Electrical Engineering,
Shanghai Jiao Tong University**

Mar 2023 – Oct 2023

Organising Committee Member

- Coordinated and assisted with the planning and execution of various youth league events.
- Collaborated with other committee members to develop event programs.

Class 2251, School of Electronic Information and Electrical Engineering

Mar 2023 – Oct 2023

Class Youth League Secretary

- Studied advanced Excel functions, enabling the handling of complex data processing tasks.
- Fostered the ability to prepare and draft official documents.
- Developed organisational and coordination skills through involvement in various activities.